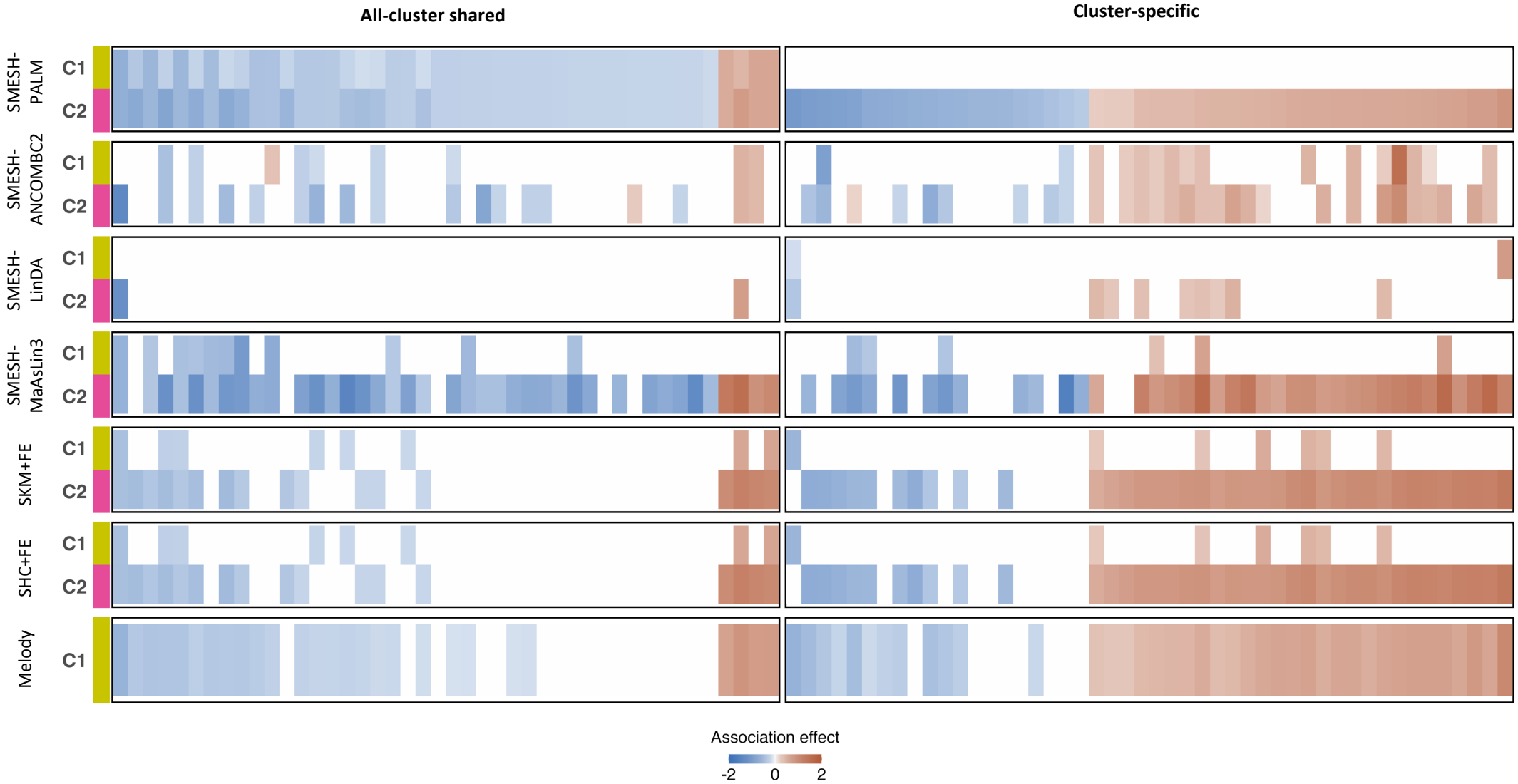
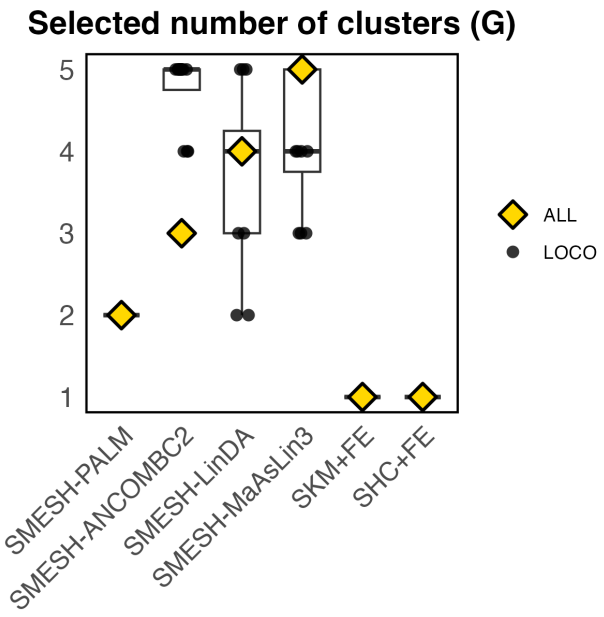
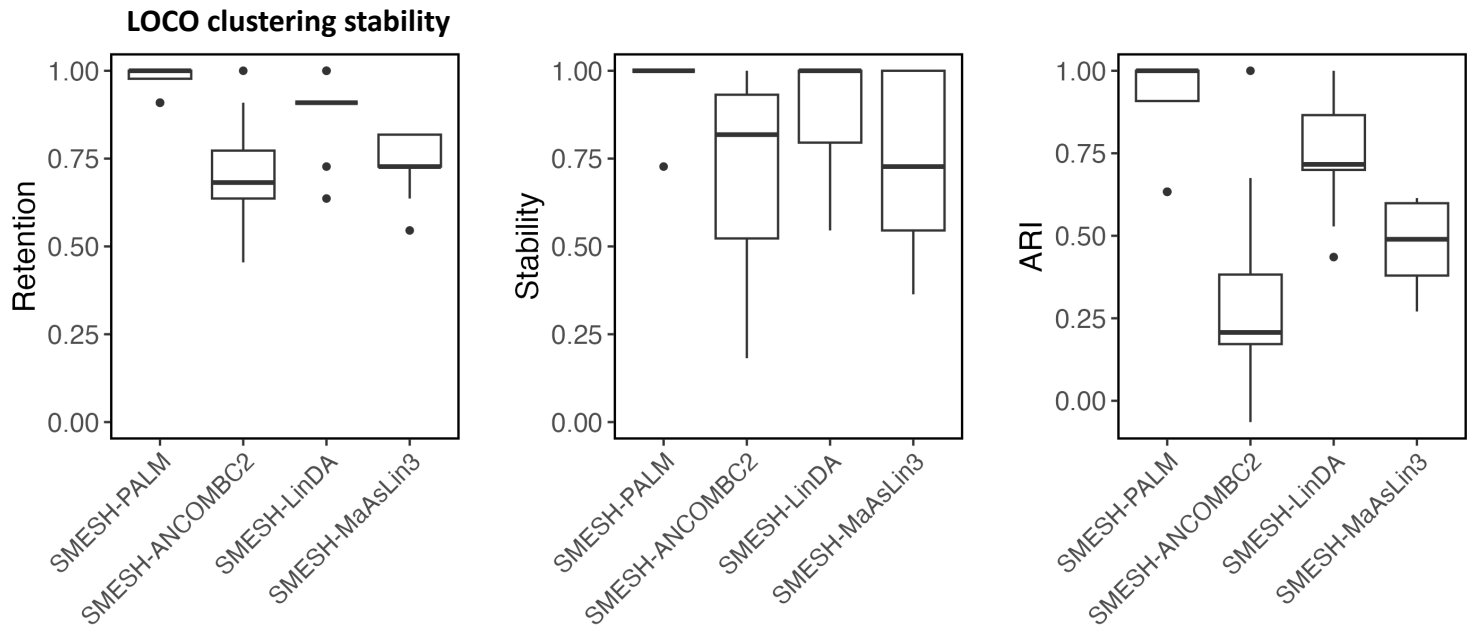
a


All-cluster shared
Cluster-specific
SMESH-PALM
SMESH-ANCOMBC2
SMESH-LinDA
SMESH-MaAsLin3
SKM+FE
SHC+FE
Melody
C1
C2
Association effect
-2
0
2

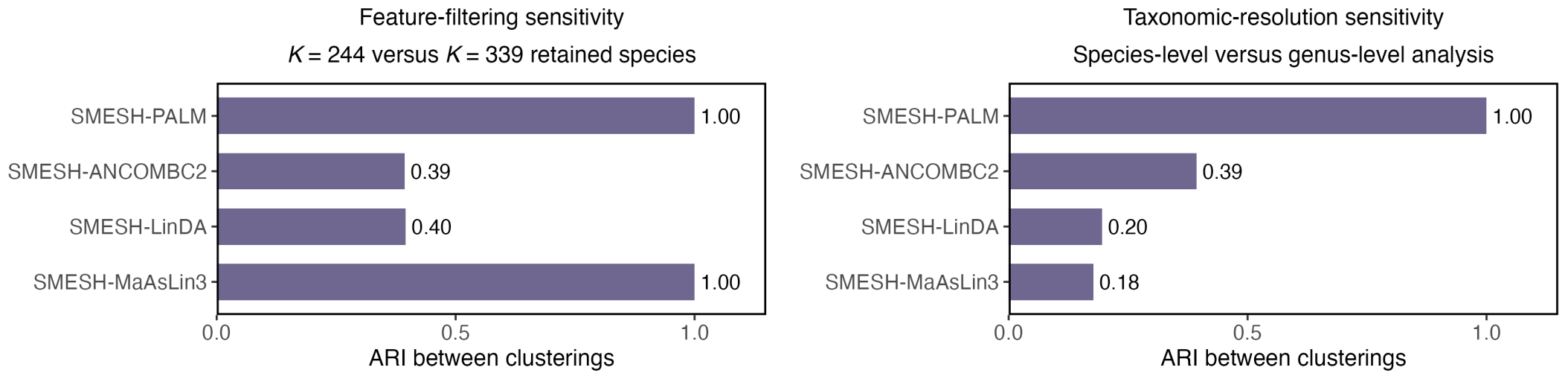
b


Selected number of clusters (G)
ALL
LOCO
SMESH-PALM
SMESH-ANCOMBC2
SMESH-LinDA
SMESH-MaAsLin3
SKM+FE
SHC+FE

c


LOCO clustering stability
Retention
Stability
ARI
SMESH-PALM
SMESH-ANCOMBC2
SMESH-LinDA
SMESH-MaAsLin3

d


Feature-filtering sensitivity
K = 244 versus K = 339 retained species
Taxonomic-resolution sensitivity
Species-level versus genus-level analysis
SMESH-PALM
SMESH-ANCOMBC2
SMESH-LinDA
SMESH-MaAsLin3
1.00
0.39
0.40
1.00
1.00
0.39
0.20
0.18
ARI between clusterings
ARI between clusterings